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A STOCHASTIC ESTIMATOR OF THE TRACE OF THE INFLUENCE MATRIX FOR LAPLACIAN SMOOTHING SPLINES

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ABSTRACT

An unbiased stochastic estimator of tr(I–A), where A is the influence matrix associated with the calculation of Laplacian smoothing splines, is described. The estimator is similar to one recently developed by Girard but satisfies a minimum variance criterion and does not require the simulation of a standard normal variable. It uses instead simulations of the discrete random variable which takes the values 1, –1 each with probability 1/2. Bounds on the variance of the estimator, similar to those established by Girard, are obtained using elementary methods. The estimator can be used to approximately minimize generalised cross validation (GCV) when using

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discretized iterative methods for fitting Laplacian smoothing splines to very large data sets. Simulated examples show that the estimated trace values, using either the estimator presented here or the estimator of Girard, perform almost as well as the exact values when applied to the minimization of GCV for n as small as a few hundred, where n is the number of data points.

1. INTRODUCTION

A popular method for fitting a surface to a set $\{z_1,..., z_n\}$ of n noisy data values, at irregularly spaced points $\{x_1,..., x_n\}$ in d–dimensional Euclidean space, is to fit a Laplacian (thin plate) smoothing spline. This may be defined as the unique function f, with suitably continuous derivatives, which minimizes

$$\sum_{i=1}^{n} (z_i - f(x_i))^2 + \rho J_m(f)$$

where $J_m(f)$ is a rotation invariant measure of the roughness of f defined in terms of m th order partial derivatives of f, with $2m > d$, and ρ is a positive number which controls the degree of data smoothing (Wahba, 1979a; Wahba and Wendelberger, 1980). The value of the smoothing parameter ρ is usually chosen to minimize the generalized cross validation (GCV) which depends on ρ. Setting $z = (z_1,..., z_n)^T$, the GCV is defined by

$$GCV = \frac{\frac{1}{n} \| (I - A)z \|^2}{[\frac{1}{n} tr(I - A)]^2}$$

where $A = (a_{ij})_{i,j=1}^{n}$ is the n x n symmetric influence matrix which takes the data values to their fitted values and $\| \cdot \|$ is the Euclidean norm (Craven and Wahba, 1979). Surfaces arising in this way depend on the data in a rotation invariant manner and exhibit near optimal convergence rates (Wahba, 1985). Their chief limitation is computational expense for large data sets. The minimum GCV univariate spline (d = 1) can be calculated

in order n operations by taking advantage of the banded nature of the associated matrices (Hutchinson and de Hoog, 1985; O'Sullivan, 1985; Hutchinson, 1986; de Hoog and Hutchinson, 1987). However, when $d > 1$, order n^3 operations are required, making Laplacian smoothing splines infeasible for data sets containing more than a few hundred points. An approximate method employing 'knots', which increases this limit to at most a few thousand data points, has been described by Bates and Wahba (1982) and implemented by Hutchinson (1984).

A practical alternative when fitting Laplacian smoothing splines to very large data sets, especially when $d = 2$, is to form a finite difference or finite element approximation to the smoothing spline (Briggs, 1974; Testud and Chong, 1983). Using a simple multi-grid iterative technique, a finite difference approximation to the 2-dimensional smoothing spline for a specified value of ρ, or even for a specified value of the residual $\|(I - A)z\|^2$, can be calculated efficiently for very large data sets (Hutchinson, 1984). However, a significant shortcoming of this approach has been the lack of a computationally feasible method for calculating $\mathrm{tr}(I-A)$ so that minimizing GCV has not been possible.

A stochastic estimator of $\mathrm{tr}(I-A)$, which can be calculated using the same technique used to calculate the smoothing spline, has recently been obtained by Girard (1987). The estimator is unbiased and has a relative standard error bounded approximately by $(2/n)^{1/2}$, making it a feasible alternative to exact calculation of $\mathrm{tr}(I-A)$ for n as small as a few hundred. This article describes another stochastic estimator of $\mathrm{tr}(I-A)$ which is distinguished from Girard's estimator chiefly in terms of its simplicity. Its form is determined by satisfying a minimum variance criterion. It is evaluated by forming the product $u^T(I-A)u = n - u^TAu$, where u is a vector of n independent samples of the discrete random variable which takes the values 1,-1 each with probability 1/2. Girard's estimator, on the other hand, uses samples of the standard normal variable and must be normalized in order to satisfy the stated bound on its relative standard error. Moreover, the bounds on the variance of the estimator presented here can be established using elementary methods.

The trace estimator is particularly appropriate in large scale regularization problems, of which data smoothing problems form a significant special case. By making reasonable assumptions on the magnitudes of the eigenvalues of A it is shown, and verified via simulation, that at least in the Laplacian smoothing spline case, the bound of $(2/n)^{1/2}$ on the relative standard error is conservative for practical values of the smoothing parameter ρ. It is also verified via simulation, for both univariate and bivariate smoothing splines, that using either the estimator presented here, or the estimator of Girard, to approximately minimize GCV, gives results that are almost as good as the exact calculation for n as small as a few hundred.

2. THE MAIN RESULT

Let A be the n x n symmetric influence matrix associated with the calculation of a Laplacian smoothing spline. The exact value of trA, and therefore of $\mathrm{tr}(I - A) = n - \mathrm{tr}A$, may be obtained by forming the sum

$$\mathrm{tr}A = \sum_{i=1}^{n} e_i^T A e_i$$

where $\{e_1,...,e_n\}$ are the n linearly independent columns of the n x n identity matrix. This is numerically impracticable, since it requires n solutions $\{Ae_1,...,Ae_n\}$ of the smoothing spline, but it does provide a motivation for the stochastic estimator considered here which is obtained by replacing these n linearly independent vectors by a single vector with *stochastically independent* entries. This reduces the computational requirement to a single solution of the smoothing spline. The precise form of the estimator is determined with the aid of the following lemma.

Lemma. Let B be an n x n symmetric matrix and let $u = (u_1,..., u_n)^T$ be a vector of n independent samples from a random variable U with mean zero and variance σ^2. Then

$$E(u^TBu) = \sigma^2 \mathrm{tr}B \qquad (1)$$

and

$$\operatorname{var}(u^T B u) = 2\sigma^4 \sum_{i \neq j} b_{ij}^2 + [E(U^4) - \sigma^4] \left(\sum_i b_{ii}^2\right). \tag{2}$$

Proof. Since

$$u^T B u = \sum_{i,j} u_i b_{ij} u_j$$

the assumptions on U show immediately that (1) holds. Using the symmetry of B it follows that

$$\begin{aligned} E[(u^T B u)^2] &= E\left[\left(\sum_{i,j} u_i b_{ij} u_j\right)\left(\sum_{r,s} u_r b_{rs} u_s\right)\right] \\ &= E\left(\sum_{i,j}\sum_{r,s} b_{ij} b_{rs} u_i u_j u_r u_s\right) \\ &= 2\sigma^4 \sum_{i \neq j} b_{ij}^2 + \sigma^4 \sum_{i \neq j} b_{ii} b_{jj} + E(U^4) \sum_i b_{ii}^2. \end{aligned}$$

Since

$$[E(u^T B u)]^2 = \sigma^4 \sum_{i,j} b_{ii} b_{jj}$$

it follows that (2) holds.

From (1) it is clear that $u^T B u$ is an unbiased estimator of trB if $\sigma^2 = 1$ and from (2) that $u^T B u$ has minimum variance if $E(U^4) - \sigma^4$ is minimized. But

$$\operatorname{var}(U^2) = E(U^2 - \sigma^2)^2 = E(U^4) - \sigma^4$$

is non–negative, with minimum value zero when U^2 is concentrated at its mean value of σ^2. This establishes the following proposition.

Proposition 1. Let B be an n x n symmetric matrix with non–zero trace. Let U be the discrete random variable which takes the values 1,–1 each with probability 1/2 and let $u = (u_1,\ldots, u_n)^T$ be a vector of n independent samples from U. Then u^TBu is an unbiased estimator of trB and

$$\text{var}(u^TBu) = 2 \sum_{i\neq j} b_{ij}^2 .$$

Moreover, U is the unique random variable amongst zero mean random variables for which u^TBu is a minimum variance, unbiased estimator of trB.

Corollary 1. Let B, U and u be as in proposition 1. Then

$$\text{var}(u^TBu) \leq 2[\text{tr}B^2 - \frac{(\text{tr}B)^2}{n}]. \tag{3}$$

Proof. Since

$$\sum_i b_{ii}^2 - \frac{(\text{tr}B)^2}{n} = \sum_i \left(b_{ii} - \frac{\text{tr}B}{n}\right)^2 \geq 0$$

it follows that

$$\sum_{i\neq j} b_{ij}^2 = \sum_{i,j} b_{ij}^2 - \sum_i b_{ii}^2 \leq \text{tr}(B^2) - \frac{(\text{tr}B)^2}{n}.$$

Corollary 2. Let U and u be as in proposition 1 and suppose that B is an n x n symmetric positive semi–definite matrix with eigenvalues lying between 0 and 1. Then

$$\text{var}(u^TBu) \leq \frac{2}{n}\text{tr}B.\text{tr}(I–B).$$

Proof. Set $B = Q^T D Q$ where Q is orthogonal and D is diagonal with entries lying between 0 and 1 (Householder, 1964). It follows that

$$trB^2 = trD^2 \leq trD = trB$$

and thence the corollary.

Corollary 3. Let A be an $n \times n$ symmetric positive semi-definite matrix with eigenvalues lying between 0 and 1. Let U and u be as in proposition 1 and suppose that $trA \leq \epsilon n$ for some $\epsilon < 1$. Then

$$\frac{var(u^T(I-A)u)}{[tr(I-A)]^2} \leq \frac{2\epsilon}{(1-\epsilon)n}.$$

In particular, if $\epsilon = 0.5$ then the relative standard error of $u^T(I-A)u$ in estimating $tr(I-A)$ is bounded by $(2/n)^{1/2}$.

Proof. Set $B = I-A$ in corollary 2 and note that the relative standard error is defined by

$$\frac{[var(u^T(I-A)u)]^{1/2}}{tr(I-A)}.$$

Remarks. If W is the standard normal variable then $E(W^2) = 1$ and $E(W^4) = 3$. If w is a vector of n independent samples of W then the lemma shows that w^TBw is an unbiased estimator of trB and $var(w^TBw) = 2\,trB^2$, as stated in théorème 4.1 of Girard (1987). Girard reduces the variance of his estimator by dividing w^TBw by w^Tw/n, to obtain a variance which is essentially the same as the right hand side of (3). From the proof of corollary 1, the estimator presented here is potentially superior to Girard's estimator if the diagonal elements of B vary significantly

about their mean value. This does not appear to be borne out by simulations for univariate and bivariate splines, but may be the case for higher dimensional splines where more data points lie on the boundary of the region occupied by data.

3. A SHARPER BOUND FOR THE VARIANCE.

The bound provided by corollary 3 is already quite practical since the influence matrix A which arises in smoothing spline problems is positive semi–definite and symmetric, with eigenvalues lying between 0 and 1. The existence of a suitably small value of ϵ is also reasonable in most smoothing spline problems where a significant amount of data smoothing is imposed. However, the simulations in the following section indicate that the bound provided by corollary 2 is conservative for small values of $\mathrm{tr}(I - A)$. By making a reasonable assumption on the distribution of the magnitude of the eigenvalues of A, corollary 1 can be used to obtain an order $n^{-1/2}$ bound for the relative standard error of $u^T(I - A)u$ in estimating $\mathrm{tr}(I - A)$ which is independent of the degree of data smoothing.

Let $M = \binom{m}{d}$ and $N = n - M$. Then, according to Wahba and Wendelberger (1980)

$$I - A = \rho V (D + \rho I)^{-1} V^T \tag{4}$$

where V is a fixed $n \times N$ matrix with orthonormal columns, D is a fixed $N \times N$ diagonal matrix with diagonal entries $(d_1,\ldots,d_N)$ given by the eigenvalues of an $N \times N$ symmetric positive definite matrix and ρ is the smoothing parameter. As $\rho \to 0$ the smoothing spline becomes an exact interpolant and $\mathrm{tr}(I - A) \to 0$. On the other hand, as $\rho \to \infty$ the smoothing spline becomes a least squares polynomial surface of degree $m - 1$ and $\mathrm{tr}(I - A) \to N$.

Wahba (1979b) has demonstrated, under mild regularity conditions on the positions of data points occupying a fixed region of d–dimensional Euclidean space, the plausibility of the conjecture that there exist constants c_1 and c_2, independent of n, such that

$$c_1 i^a \leq n/d_i \leq c_2 i^a \tag{5}$$

for $i = 1,\ldots, N$ where $a = 2m/d$. Utreras (1983) has proved this conjecture in the case of equi–spaced data with $d=1$.

Proposition 2. Let $B = I - A$ in (4) and assume that (5) holds. Set $c = c_2/c_1$ and let U and u be as in proposition 1. Then

$$\frac{\mathrm{var}(u^T B u)}{(\mathrm{tr}B)^2} \leq \frac{2}{N}\left[\frac{c^2 a^2}{2a+1} + c^2 - 1\right].$$

Proof. From corollary 1,

$$\frac{\mathrm{var}(u^T B u)}{2(\mathrm{tr}B)^2} \leq \frac{\mathrm{tr}B^2}{(\mathrm{tr}B)^2} - \frac{1}{n} \leq \frac{\mathrm{tr}B^2}{(\mathrm{tr}B)^2} - \frac{1}{N}$$

so it remains to find a bound for

$$T = \frac{\mathrm{tr}B^2}{(\mathrm{tr}B)^2}.$$

Setting $a_i = 1/(\rho + d_i)$ for each i, it follows from (4) that

$$T = \frac{\Sigma\, a_i^2}{(\Sigma\, a_i)^2}. \tag{6}$$

Since $\dfrac{da_i}{d\rho} = -a_i^2$, it then follows that

$$\frac{dT}{d\rho} = \frac{-2}{(\Sigma a_i)^3}\left[(\Sigma a_i)(\Sigma a_i^3) - (\Sigma a_i^2)(\Sigma a_i^2)\right]$$

$$= \frac{-2 \sum_{i,j} a_i^2 a_j (a_i - a_j)}{(\Sigma\, a_i)^3}$$

$$= \frac{-\sum_{i,j} a_i\, a_j (a_i - a_j)^2}{(\sum a_i)^3}$$

so that T is a decreasing function of ρ which attains its maximum as $\rho \to 0$. Note that this does not depend on assumption (5). Setting $\rho = 0$ in (6) and finally making use of (5) then gives

$$T \leq \frac{\sum 1/d_i^2}{(\sum 1/d_i)^2} \leq \frac{c^2 \sum_1^N i^{2a}}{(\sum_1^N i^a)^2} \leq \frac{c^2 (a+1)^2}{(2a+1)\, N}$$

from which the proposition follows.

Remark. Numerical evidence presented by Wahba (1979b) suggests that, except for the few largest eigenvalues, (5) holds with c_1 quite close to c_2 so, for most practical purposes, proposition 2 can be assumed to hold with $c = 1$.

4. AN APPROXIMATE BOUND FOR THE VARIANCE FOR ARBITRARY ρ

The bound provided by proposition 2, with $c = 1$, is still somewhat conservative because ρ does not approach zero in most smoothing spline problems. An approximate bound on the variance for arbitrary ρ is now presented which is valid provided $\mathrm{tr}(I-A)$ is not close to zero. The bound depends on making integral approximations for $\mathrm{tr}B$ and $\mathrm{tr}B^2$.

Set $B = I - A$ in (4) and assume that (5) holds. Then

$$\mathrm{tr}B = \sum_{i=1}^N \frac{\rho}{\rho + d_i} = \sum_{i=1}^N \frac{c_3 \rho i^a}{1 + c_3 \rho i^a}$$

for some c_3 which depends on ρ and satisfies $c_1/n \leq c_3 \leq c_2/n$. An integral approximation for $\mathrm{tr}B$ is then given by

$$\begin{aligned} \mathrm{trB} &= \int_0^N \frac{c_3\rho x^a\,dx}{1+c_3\rho x^a} + O(1) \\ &= N - \int_0^N \frac{dx}{1+c_3\rho x^a} + O(1) \\ &= N\,(1 - f_a(y_3)/y_3) + O(1) \end{aligned} \tag{7}$$

where $y_3 = (c_3\rho)^{1/a} N$ and

$$f_a(y) = \int_0^y \frac{dt}{1+t^a}.$$

Similarly,

$$\mathrm{trB}^2 = \sum_{i=1}^{N} \frac{\rho^2}{(\rho+d_i)^2} = \sum_{i=1}^{N} \frac{(c_4\rho i^2)^2}{(1+c_4\rho i^2)^2}$$

for some c_4 satisfying $c_1/n \le c_4 \le c_2/n$. Using the identity

$$\frac{x^2}{(1+x)^2} = 1 - \frac{1}{1+x} - \frac{x}{(1+x)^2}$$

an integral approximation for trB^2 is then given by

$$\mathrm{trB}^2 = N - \int_0^N \frac{dx}{1 + c_4\rho x^a} - \int_0^N \frac{c_4\rho x^a\,dx}{(1 + c_4\rho x^a)^2} + O(1)$$

Integrating by parts gives

$$\int_0^N \frac{dx}{1 + c_4\rho x^a} = \frac{N}{1 + c_4\rho N^a} + a\int_0^N \frac{c_4\rho x^a\,dx}{(1 + c_4\rho x^a)^2}$$

so that

$$trB^2 = N - N[(a+1)/a][f_a(y_4)/y_4] - N/[a(1+y_4^a)] + O(1)$$

where $y_4 = (c_4\rho)^{1/a}N = (c_4/c_3)^{1/a}y_3$. Ignoring the $O(1)$ terms, permissible provided both trB and trB^2 significantly exceed 1, and using corollary 1, an approximate bound for the variance is then given by

$$\frac{var(u^TBu)}{(trB)^2} \lesssim (\frac{2}{N})\ g_a(y_3, y_4) \tag{8}$$

where

$$g_a(y_3, y_4) = \frac{1 - [(a+1)/a][f_a(y_4)/y_4] - 1/[a(1+y_4^a)]}{[1 - f_a(y_3)/y_3]^2} - 1$$

and y_3, y_4 are as above.

The bound given by (8) has been verified by simulation for the two most common cases of univariate (d=1) and bivariate (d=2) smoothing splines with m=2. For these cases the function f_a can be calculated explicitly. The solid curve in Figure 1 shows $g_a(y_3, y_4)$ as a function of trB/N, via approximation (7), with $y_4=y_3$, for the univariate spline (d=1, m=2, a=4). The lower and upper dashed curves show $g_a(y_3, y_4)$, also as a function of trB/N, with $y_4=0.975y_3$ and $y_4=1.025y_3$ respectively. The heights of the curves at $trB=0$ are given by the values of $c^2a^2/(2a+1) + c^2 - 1$, as afforded by proposition 2, with $c = (y_4/y_3)^a$. The points plotted in Figure 1 correspond to values of $(\frac{N}{2})\ \frac{var(u^TBu)}{(tr\ B)^2}$ estimated via simulation for an equispaced data set and a non–equispaced data set, each with $n = 500$. The equispaced data set was obtained by setting

$$x_i = i \quad (i = 1,..., 500)$$

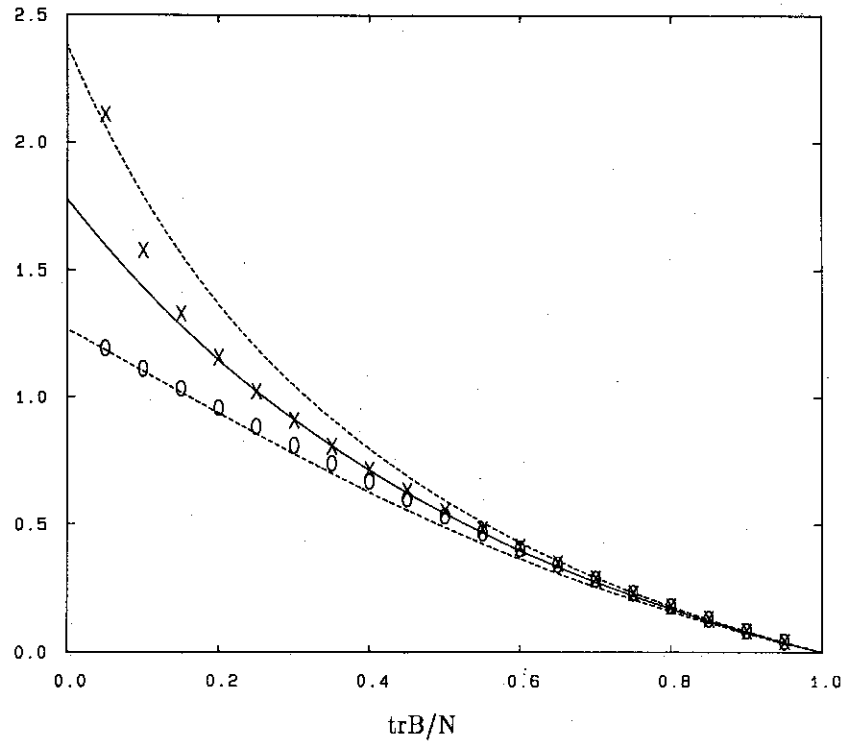


FIG. 1 Estimated values of $(\frac{N}{2})\ \frac{var(u^TBu)}{(trB)^2}$ as a function of trB/N for the univariate smoothing spline. Simulated values for equispaced and non–equispaced data sets are indicated by O , X repectively.

and the non–equispaced data set by setting

$$x_i = i + 0.99\ v_i \quad (i = 1,..., 500) \tag{9}$$

where each v_i was an independent sample from the uniform random variable on the interval $[-1/2, 1/2]$. For each data set, the vector u was simulated 1000 times in order to calculate an estimate of $var(u^TBu)$ for each value of trB. The relative standard errors of the plotted point values are less than 5%. The value of ρ corresponding to each value of trB and the values of

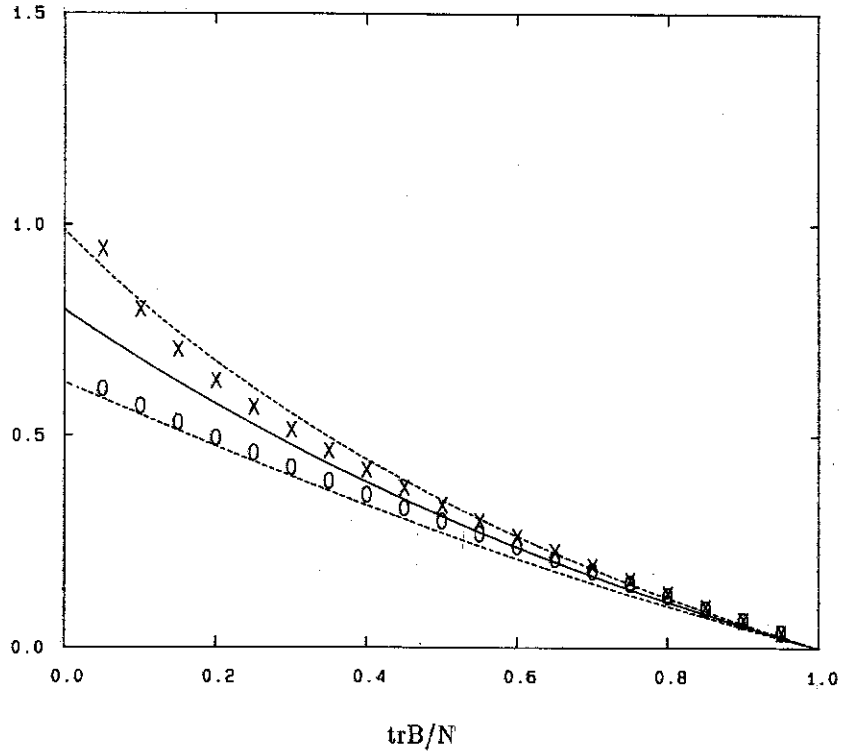


FIG. 2 Estimated values of $(\frac{N}{2})\ \frac{var(u^T Bu)}{(trB)^2}$ as a function of trB/N for the bivariate smoothing spline. Simulated values for equispaced and non–equispaced data sets are indicated by O , X respectively.

$u^T Bu$ for each simulation were calculated using a modification of subroutine CUBGCV (Hutchinson, 1986).

Figure 2 shows the corresponding curves and simulated point values for the bivariate spline (d=2, m=2, a=2). The values of y_3, y_4 were specified as for the univariate example. The point values were obtained by simulating the vector u 500 times for both an equispaced and a non–equispaced data set with n = 256. The equispaced data set consisted of points on a 16 x 16 square lattice with unit spacing in the x and y directions. The

non–equispaced data set was obtained by perturbing independently the x and y coordinates of the equispaced data set in the same manner as in the univariate example. The relative standard errors of the plotted point values are approximately 6%. Values of ρ and $u^T Bu$ were calculated in this case using routines from the procedure SPLIN2A (Hutchinson, 1984).

In both the univariate and bivariate cases, the bound provided by (8), with $y_4 = y_3$, is slightly conservative for the regularly spaced data sets but satisfactory for the irregularly spaced data for all but the smallest values of trB/N. Thus, a bound of $(2/N)^{1/2}$ on the relative standard deviation of $u^T Bu$ in estimating trB appears to be acceptable provided trB/N exceeds 0.3 in the univariate case and 0.1 in the bivariate case. The dashed lines in Figures 1, 2 give an indication of the range of variation of y_4/y_3, and hence of c_4/c_3, for practical values of the smoothing parameter.

5. MINIMIZING GCV

The crucial test of the trace approximation is its performance in minimizing GCV. Noisy data values were associated with the irregularly spaced position data sets of the previous section. Each noisy data set in the univariate case consisted of a set of values

$$z_i = \sin(2\pi x_i/t) + \epsilon_i \quad (i=1,\ldots, 500)$$

where the x_i were given by (9) and each ϵ_i was an independent sample from a normal variable with mean zero and variance 0.1^2. The parameter t successively took on values ranging from 10 to 250, corresponding to decreasing complexity of the true function. Table I shows the results, in terms of root mean square residuals of the fitted values from the true values, of minimizing GCV using in turn the exact value of trB, the stochastic estimate of trB presented here and Girard's estimate of trB. The simulations were replicated 25 times for each value of t by holding the x_i fixed but using different simulations of both the ϵ_i and the random vector u used to calculate the estimates of trB.

Table II shows the corresponding results in the bivariate case where noisy data values were simulated for each value of t by setting

Table I. Root mean square residuals of 500 smoothed univariate noisy data values from the true values obtained by minimizing exact and approximate values of GCV for 25 replicates for each value of t. The mean value of trB/N for the exact method is also tabulated.

	Exact GCV		Hutchinson's approximation	Girard's approximation
t	Root mean square residual	Mean trB/N	Root mean square residual	Root mean square residual
10	0.0678	0.474	0.0686	0.0696
25	0.0460	0.763	0.0461	0.0463
100	0.0255	0.929	0.0259	0.0257
250	0.0180	0.969	0.0200	0.0185
500	0.0147	0.983	0.0183	0.0161

Table II. Root mean square residuals of 256 smoothed bivariate noisy data values from the true values obtained by minimizing exact and approximate values of GCV for 25 replicates for each value of t. The mean value of trB/N for the exact method is also tabulated.

	Exact GCV		Hutchinson's approximation	Girard's approximation
t	Root mean square residual	Mean trB/N	Root mean square residual	Root mean square residual
8	0.0743	0.319	0.0761	0.0785
12	0.0556	0.599	0.0560	0.0567
16	0.0480	0.706	0.0485	0.0490
32	0.0329	0.868	0.0338	0.0343
48	0.0265	0.912	0.0277	0.0285

$$z_{ij} = \sin(2\pi x_i/t).\sin(2\pi y_j/t) + \epsilon_{ij} \qquad (i, j=1,..., 16)$$

where the x_i, y_i were the coordinates of the irregularly spaced bivariate position data of the previous section and each ϵ_{ij} was an independent sample from a normal variable with mean zero and variance 0.1^2.

The approximate methods are in good agreement with the exact method in both the univariate and bivariate cases. In view of the order

$n^{-1/2}$ bound on the relative standard error of the stochastic estimates of the trace, the performance of both approximate methods could be expected to be even closer to that of the exact method for larger values of n. There appears to be little practical difference in the accuracy of the two stochastic estimators. The addition of an approximate GCV minimization routine, based on the stochastic trace approximation presented here, to an iterative discretized bivariate smoothing spline procedure suitable for large data sets will be the subject of a separate study.

BIBLIOGRAPHY

Bates, D. and Wahba, G. (1982). Computational methods for generalized cross validation with large data sets. In *Treatment of Integral Equations by Numerical Methods* (C.T.H. Baker and G.F. Miller, eds.), 283–296. New York : Academic Press.

Briggs, I.C. (1974). Machine contouring using minimum curvature. *Geophysics* **39**, 39–48.

Craven, P. and Wahba, G. (1979). Smoothing noisy data with spline functions. *Numer. Math.* **31**, 377–403.

Girard, D. (1987). Un algorithme simple et rapide pour la validation croissee generalisee sur des problemes de grande taillee. RR 669–M, Grenoble, France: Informatique et Mathématiques Appliquées de Grenoble.

de Hoog, F.R. and Hutchinson, M.F. (1987). An efficient method for calculating smoothing splines using orthogonal transformations. *Numer. Math.* **50**, 311–319.

Householder, A.S. (1964). *The Theory of Matrices in Numerical Analysis.* New York: Blaisdell.

Hutchinson, M.F. (1984). A summary of some surface fitting and contouring programs for noisy data. CSIRO Division of Mathematics and Statistics Counsulting Report, ACT 84/6, Canberra.

Hutchinson, M.F. (1986). A fast procedure for calculating minimum cross validation cubic smoothing splines. *ACM Trans. Math. Softw.* **12**, 150–153.

Hutchinson, M.F. and de Hoog, F.R. (1985). Smoothing noisy data with spline functions. *Numer. Math.* **47**, 99–106.

O'Sullivan, F. (1985). Comments (pp 39–40) in: Silverman, B.W. (1985). Some aspects of the spline smoothing approach to non–parametric regression curve fitting. *J. R. Statist. Soc. B.* **47**, 1–52.

Testud, J. and Chong, M. (1983). Three–dimensional wind field analysis from dual–doppler radar data. Part I: Filtering and differentiating the raw data. *J. Climate Appl. Meteorol.* **22**, 1204–1215.

Utreras, F. (1983). Natural spline functions their associated eigenvalue problem. *Numer. Math.* **42**, 107–117.

Wahba, G. (1979a). How to smooth curves and surfaces with splines and cross validation. *Proc. 24th Design of Experiments Conference, U.S. Army Research Office Rep.* 79–2, 167–192.

Wahba, G. (1979b). Convergence rates of "Thin Plate" smoothing splines when the data are noisy. In *Smoothing Techniques for Curve Estimation* (T. Gasser and M. Rosenblatt, eds.), *Lecture Notes in Mathematics No. 757.* Berlin: Springer.

Wahba, G. (1985). A comparison of GCV and GML for choosing the smoothing parameter in the generalized spline smoothing problem. *Ann. Statist.* **13**, 1378–1402.

Wahba, G. and Wendelberger, J. (1980). Some new mathematical methods for variational objective weather analysis using splines and cross–validation. *Mon. Weath. Rev.* **108**, 1122–1143.

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